Stretches

Exam code: 9MA0

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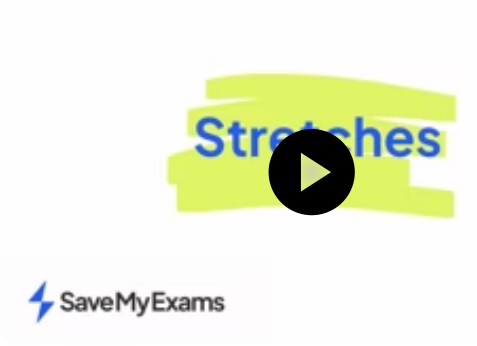

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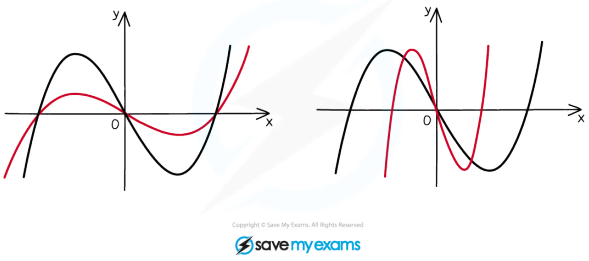
Stretches

What are graph stretches?

- When you alter a function in certain ways, the effects on the graph of the function can be described by geometrical transformations
- With a **stretch** all the points on the graph are moved towards or away from either the ***x*** or the ***y*** axis by a constant scale factor



What do I need to know about graph stretches?

- The graph of ***y* = *af*(*x*)** is a **vertical** stretch of the graph ***y* = *f*(*x*)** by a scale factor of ***a***, centred on the ***x*** axis

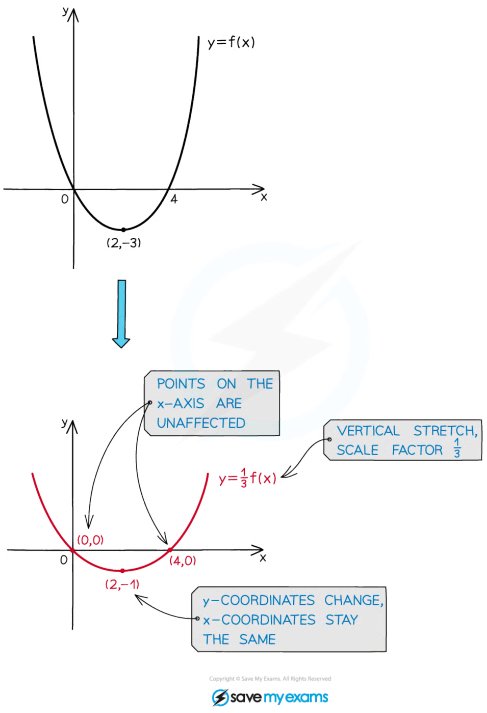
Help me understand

Stuck on something?

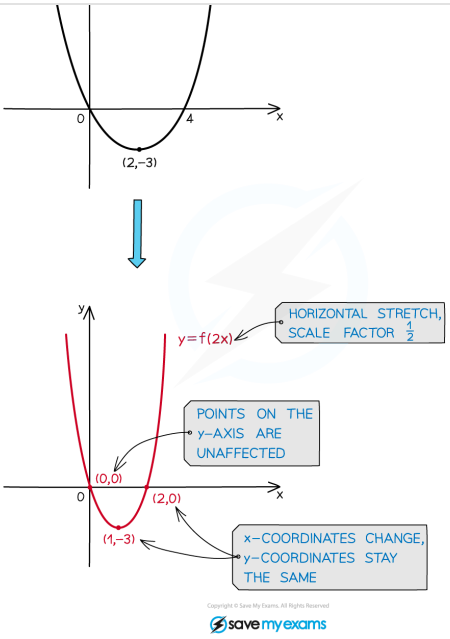
Ask questions and get help with exactly what you're revising.

Got it

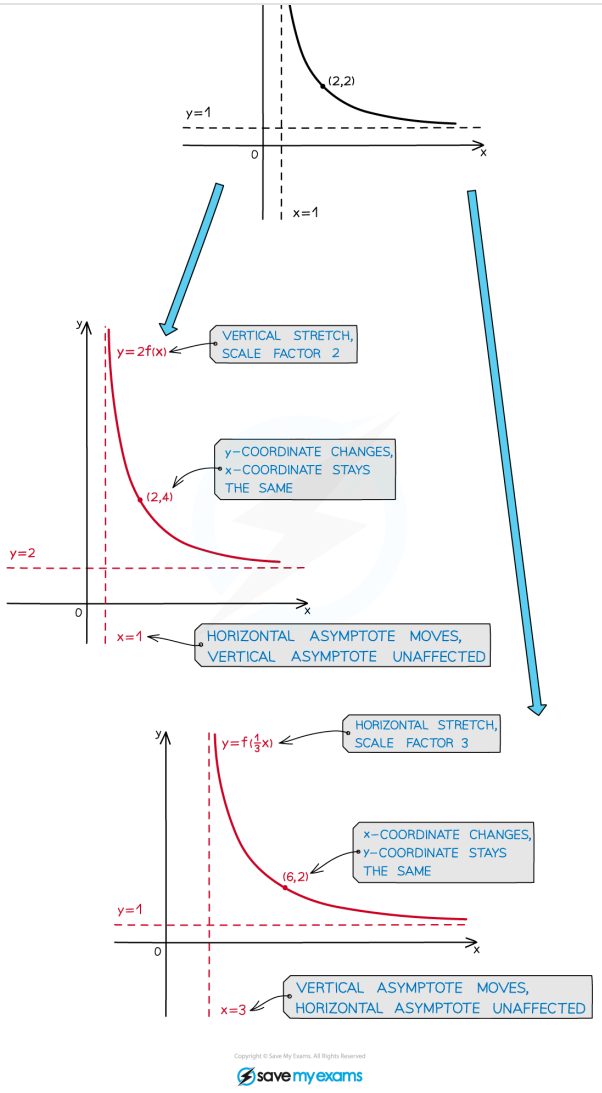
- All other points move **parallel to the *y* axis**, away from (***a* > 1**) or towards (**0 < *a* < 1**) the *x* axis



- The graph of ***y* = f(*ax*)** is a **horizontal** stretch of the graph ***y* = f(*x*)** by a scale factor of $\frac{1}{a}$, centred on the ***y*** axis
  - The ***y*** coordinates of points stay the same; ***x*** coordinates are multiplied by $\frac{1}{a}$
  - Points **on the *y* axis** stay where they are
  - All other points move **parallel to the *x* axis**, away from (**0 < *a* < 1**) or towards (***a* > 1**) the *y* axis



- Any asymptotes of **f(x)** are also affected by the stretch (stretch them as you would stretch the function of a straight line)
- If an asymptote is one of the coordinate axes, or is parallel to the direction of the stretch, however, it will not be affected



Examiner Tips and Tricks

- When you sketch a stretched graph, be sure to indicate the new coordinates of any points that are marked on the original graph.
- Try to indicate the coordinates of points where the stretched graph intersects the coordinate axes (if you don't have the equation of the original function this may not be possible).
- If the graph has asymptotes, don't forget to sketch the asymptotes of the stretched graph as well.

Worked Example

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